



DIPARTIMENTO DI ELETTRONICA,
INFORMAZIONE E BIOINGEGNERIA

Politecnico di Milano

Machine Learning (Code: 097683)

February 7, 2017

Name:

Surname:

Student ID:

Row:

Column:

Time: 2 hours 30 minutes

Prof. Marcello Restelli

Maximum Marks: 31

- The following exam is composed of **10 exercises** (one per page). The first page needs to be filled with your **name, surname and student ID**. The following pages should be used **only in the large squares** present on each page. Any solution provided outside these spaces will not be considered for the final mark.
- During this exam you are **not allowed to use electronic devices** like laptops, smartphones, tablets and/or similar. As well, you are not allowed to bring with you any kind of note, book, written scheme and/or similar. You are also not allowed to communicate with other students during the exam.
- The first reported violation of the above mentioned rules will be annotated on the exam and will be considered for the final mark decision. The second reported violation of the above mentioned rules will imply the immediate expulsion of the student from the exam room and the **annulment of the exam**.
- You are allowed to write the exam either with a pen (black or blue) or a pencil. It is your responsibility to provide a readable solution. We will not be held accountable for accidental partial or total cancellation of the exam.
- The exam can be written either in **English** or **Italian**.
- You are allowed to withdraw from the exam at any time without any penalty. You are allowed to leave the room not early than half the time of the duration of the exam. You are not allowed to keep the text of the exam with you while leaving the room.

Ex. 1	Ex. 2	Ex. 3	Ex. 4	Ex. 5	Ex. 6	Ex. 7	Ex. 8	Ex. 9	Ex. 10	Tot.
/ 5	/ 5	/ 5	/ 2	/ 2	/ 2	/ 2	/ 2	/ 3	/ 3	/ 31

Exercise 1 (5 marks)

Describe and compare ridge regression and lasso.

Exercise 2 (5 marks)

Define the VC dimension and describe the importance and usefulness of VC dimension in machine learning.

Exercise 3 (5 marks)

Describe the policy iteration algorithm.

Exercise 4 (2 marks)

You are fitting a Cubic to your data set, i.e., a Linear Model with cubic features $\phi(x) = [1, x, x^2, x^3]$. You are thinking to use only linear features ($\phi(x) = [1, x]$). Which of the following is most likely true:

1. Using the Linear Model will decrease the sum of Bias and Variance.
2. Using the Linear Model will decrease the Irreducible Error;
3. Using the Linear Model will decrease the Bias of your model;
4. Using the Linear Model will decrease the Variance of your model;

Provide motivations to your answers.

1. TRUE/FALSE: in the case we are using a model which is more adequate to explain the phenomenon generating the considered data, we will reduce the sum of Bias and Variance. Vice versa the sum of the twos might increase.
2. FALSE: the Irreducible Error is related to the noise intrinsic to the data and, thus, can not be reduced by changing the model.
3. FALSE: the more the model is simple, the more it is likely that it has some bias w.r.t. the real process generating data.
4. TRUE: if we have a simpler model we reduce the model Variance.

Exercise 5 (2 marks)

Consider the following statements. Tell whether they are true or false and provide motivation for your answers.

1. K-means and GMM consider all the dimensions of the dataset differently;
2. GMM defines a probability of inclusion of a sample in a cluster;
3. Both GMM and K-means are not sensitive to outliers;
4. K-means is an Expectation-Maximization algorithm.

1. FALSE: GMM considers differently each dimension, by considering a generic covariance matrix while estimating the Gaussian distributions.
2. TRUE: The likelihood of a sample defines also an un-normalized version of the probability that a sample belongs to a cluster.
3. FALSE: K-means is highly influenced by outliers. Sometimes, during the optimization procedure, clusters containing a single outlier might be created.
4. TRUE: it is a special case of the EM algorithm, where the E step consists in the inclusion of data into clusters and the M step consists in the update of the centroids.

Exercise 6 (2 marks)

Which of the following statements are true?

1. If you are training multi-class SVM with the one-vs-one method, it is not possible to use a kernel.
2. The maximum value of the Quadratic kernel $K(\mathbf{x}_i, \mathbf{x}_j) = (1 + \mathbf{x}_i^T \mathbf{x}_j)^2$ inside a unitary ball with norm 2 ($\|\mathbf{x}\|_2 \leq 1$) is 1.
3. If the data are linearly separable, an SVM using a linear kernel will return the same parameters w regardless of the chosen value of C .
4. Suppose you have 3D input examples (i.e., $\mathbf{x}_i \in \mathbb{R}^3$). The decision boundary of the SVM (with the linear kernel) is a 2D plain.

1. FALSE: the use of the kernel does not influence the method we consider to handle multiple classes.
2. FALSE: $K(\mathbf{x}, \mathbf{x}) = 1 + \|\mathbf{x}\|_2^2 \geq 1$.
3. FALSE: the parameter C influences the thickness of the intra-margin region, thus also the decision boundary position.
4. TRUE: linear SVM considers a decision boundary in a $d - 1$ dimension, where d is the input dimension.

Exercise 7 (2 marks)

Tell whether the following problems can be modeled as MAB and explain why?

1. Web ads display
2. Play “Space Invaders”
3. Weather prediction (with multiple experts)
4. Maze escape

If they fit the MAB setting, is the environment adversarial or stochastic?

1. MAB, stochastic: we can select a single ad at each time point and receive a feedback (click/no-click) from a user, sampled from a user distribution.
2. MDP: the state is not unique and the optimal action is dependent on the game state.
3. Expert: we receive feedback from all the considered experts. If we consider a feedback coming only from the chosen expert it is a MAB (stochastic).
4. MDP: the optimal action we might perform in a maze is dependent on the position we are in.

Exercise 8 (2 marks)

Suppose you have trained a logistic regression classifier l on a dataset $Z = \{(x_n, t_n)\}_n$ and the output corresponding to observation x_n is $\hat{y}_n$. Currently, you predict class 1 if $\hat{y}_n > \tau$, and predict 0 if $\hat{y}_n < \tau$, with $\tau = 0.5$. Suppose you increase the threshold to $\tau = 0.8$ getting a new classifier l_{new} . Which of the following are true? Check all that apply and provide a motivation.

1. The number of samples x_n from Z classified as positive instance will decrease if we use l_{new} instead of l .
2. The number of samples x_n from a test dataset Z_{test} classified as positive instance will decrease if we use l_{new} instead of l .
3. The classifier l_{new} is likely to have a higher accuracy.
4. The classification error over Z might decrease by using l_{new} instead of l .

1. TRUE: we are shrinking the positive region, thus it might happen that some points which were classified as positive will be classified as negative with l_{new} .
2. TRUE: same reason as in 1.
3. TRUE/FALSE: the amount of true positive increases and the amount of true negative decreases, thus it might happen that the accuracy increases or decreases.
4. FALSE: if we properly trained the logistic regression the classification error should increase when we use a sub-optimal decision boundary.

Exercise 9 (3 marks)

Consider an initial parameter $\mathbf{w}^{(0)} = [0 \ 0.5 \ 0.2]$ and a loss function of the form:

$$J(\mathbf{w}) = \frac{1}{2N} \sum_n -t_n \log(\mathbf{x}_n^T \mathbf{w}) - (1 - t_n) \log(1 - \mathbf{x}_n^T \mathbf{w}).$$

Derive the update given from the gradient descent for the datum $x_1 = [1 \ 3 \ 2]$, $t_1 = 1$ and a learning rate $\alpha = 0.3$.

Do you think the prediction $\mathbf{x}_n^T \mathbf{w}$ would be appropriate for a binary classification problem? Motivate your answers.

$$\begin{aligned} \frac{\partial J(\mathbf{w})}{\partial w_i} &= \frac{1}{2N} \sum_n \left(-\frac{x_{ni} t_n}{\mathbf{x}_n^T \mathbf{w}} + \frac{(1 - t_n) x_{ni}}{1 - \mathbf{x}_n^T \mathbf{w}} \right) \\ &= \frac{1}{2N} \sum_n \frac{x_{ni} (\mathbf{x}_n^T \mathbf{w} - t_n)}{\mathbf{x}_n^T \mathbf{w} (1 - \mathbf{x}_n^T \mathbf{w})} \end{aligned}$$

In the case we have a single data:

$$\frac{\partial J(\mathbf{w})}{\partial w_i} = \frac{x_{1i}}{2} \frac{(\mathbf{x}_1^T \mathbf{w} - t_1)}{\mathbf{x}_1^T \mathbf{w} (1 - \mathbf{x}_1^T \mathbf{w})} \rightarrow \nabla J(\mathbf{w}) = \frac{\mathbf{x}_1}{2} \frac{(\mathbf{x}_1^T \mathbf{w} - t_1)}{\mathbf{x}_1^T \mathbf{w} (1 - \mathbf{x}_1^T \mathbf{w})}$$

and the parameter update given by the gradient descent would be:

$$\mathbf{w}^{(k)} \leftarrow \mathbf{w}^{(k-1)} - \alpha \nabla J(\mathbf{w}).$$

For the specific datum we have:

$$\mathbf{w}^{(k)} = \begin{bmatrix} 0 \\ 0.5 \\ 0.2 \end{bmatrix} + \frac{3}{10} \frac{1}{2} \frac{10}{19} \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 0.078 \\ 0.736 \\ 0.357 \end{bmatrix}$$

The prediction would not be appropriate since the product $\mathbf{x}_n^T \mathbf{w}$ might be outside the interval $[0, 1]$ and in this case the prediction would not make sense.

Exercise 10 (3 marks)

Consider the MDP with 3 states $\mathcal{S} = \{A, B, C\}$ (C is terminal), 2 actions $\mathcal{A} = \{h, l\}$ common to all states. Policy π generated the following trajectories (expressed as state/action/reward tuples):

$(A, h, 3) \rightarrow (B, r, 2) \rightarrow (C)$

$(A, h, 2) \rightarrow (B, h, 1) \rightarrow (C)$

$(B, r, 1) \rightarrow (A, h, 1) \rightarrow (C)$

1. Can you tell, without computing anything, if by resorting to MC with every-visit and first-visit approach you will have different results?
2. Compute the values with the two aforementioned methods.
3. Assume to consider a discount factor $\gamma = 1$. Compute the values by resorting to TD? Assume to start from zero values for each state and $\alpha = 0.1$.

Provide the adequate formulas to justify your computations.

1. The twos would give the same result since there are no repeated states in single trajectory.

2.

$$V(A) = \frac{5 + 3 + 1}{3} = 3$$

$$V(B) = \frac{2 + 1 + 2}{3} = \frac{5}{3}$$

3.

$$V(A) \leftarrow V(A) + \alpha(r + \gamma V(B) - V(A)) = 0 + 0.1(3 + 0 - 0) = 0.3$$

$$V(B) \leftarrow V(B) + \alpha(r + \gamma V(C) - V(B)) = 0 + 0.1(2 + 0 - 0) = 0.2$$

$$V(A) = 0.3 + 0.1(2 + 0.2 - 0.3) = 0.49$$

$$V(B) = 0.2 + 0.1(1 + 0 - 0.2) = 0.28$$

$$V(B) = 0.28 + 0.1(1 + 0.49 - 0.28) = 0.401$$

$$V(A) = 0.49 + 0.1(1 + 0 - 0.49) = 0.541$$